Ministry of Higher Education
and Scientific Research

Syrian Private University

Faculty of Engineering

Department of Computer
Systems and Network

Graduation Project Title:

Development of a Hybrid System (IDS & Honeypot) for Securing the Physical Layer in IoT Networks Against Signal Leakage Attacks and Smart Jamming

2025/2026

Prepared by:

Qasim Oqlah

Saad Al-Mubarak

Samer Rahma

Supervised by:

Dr.Waseem Junaidi

Dr.Yamen Al-Hallak

SUPERVISOR CERTIFICATION

I certify that the preparation of this project entitled

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………………………………………….. , It was carried out under my supervision in the Department of Computer Systems and Network Security – Faculty of Engineering, and it is part of the requirements for obtaining a degree in Computer Systems and Network Security Engineering.

Contents

Abstract:

The exponential proliferation of Internet of Things (IoT) devices across critical infrastructure domains has introduced severe security vulnerabilities at the physical layer, where conventional cryptographic mechanisms are fundamentally incapable of addressing radio-frequency threats such as Signal Leakage and Smart Jamming. This paper presents a novel hybrid security framework that integrates a Tiny Machine Learning Intrusion Detection System (TinyML-IDS) with an automated Honeypot deception layer, forming a unified pipeline specifically engineered to detect, classify, and actively mitigate physical layer attacks in smart home IoT networks in real time.

The IDS component employs a lightweight One-Dimensional Convolutional Neural Network (1D-CNN) trained on a real-world dataset of 1,230,000 Received Signal Strength (RSS) measurements collected via GNU Radio software-defined radio hardware, spanning three attack scenarios: Normal communication, Constant Jamming, and Smart Jamming. The model achieves an overall classification accuracy of 97.11%, a weighted F1-Score of 97.10%, and a mean inference latency of 31.3 milliseconds, confirming its suitability for real-time edge deployment under stringent TinyML resource constraints. The architecture incorporates Gaussian noise augmentation, Mixup regularization, Dropout, and L2 weight decay to ensure robust generalization across unseen signal conditions.

The Honeypot component, implemented using the Cowrie medium-interaction framework, operates as an active defense layer that is automatically triggered by the IDS upon detecting three consecutive attack packets. Upon activation, the system redirects adversarial traffic toward purpose-configured emulated IoT decoy devices — a Camera node and a SmartDoor node — without exposing the actual network infrastructure, while simultaneously logging adversarial behavioral intelligence for forensic analysis.

The complete framework is evaluated within a dual-simulation environment comprising a ten-node smart home network modeled in NS-3 with NetAnim visualization, operating across Wi-Fi (IEEE 802.11b, 2.4 GHz) and LoRa (868 MHz) protocols, and a physical layer signal generator implemented in GNU Radio 3.10.9. A multi-layer feature fusion strategy combines physical layer RSS measurements with network layer features extracted from NS-3 PCAP traces — including packet loss rate, transmission delay, retry count, and packet size — yielding a richer feature representation than single-layer approaches. The integrated hybrid pipeline successfully activates both Honeypot devices across distinct attack scenarios, achieving a system-level response latency of 46.7 milliseconds. These results collectively demonstrate that the proposed framework delivers high-accuracy threat detection, automated active defense, and adversarial intelligence collection within a

single cohesive architecture, addressing a critical gap in the existing IoT security literature where IDS and Honeypot mechanisms have predominantly been developed and evaluated in isolation.

Introduction:

In recent years, the Internet of Things (**IoT**) has experienced tremendous growth. Billions of smart devices such as sensors, cameras, wearable devices, and industrial systems are connected through the Internet.

Estimates indicate that the number of IoT devices may exceed 80 billion devices in the coming years, making this technology essential for smart cities, industry healthcare and intelligent transportation.

IoT networks rely heavily on wireless communications to transmit data between devices and nodes. While this provides flexibility, it also exposes the network to many security threats, especially at the physical layer.

Attackers may intercept signals, jam communications, or exploit wireless channel characteristics.

The **physical layer** is responsible for transmitting electromagnetic signals through the wireless medium.

Since signals propagate through the air, any device within range may capture or interfere with them.

This leads to attacks such as eavesdropping, jamming and spoofing.

The danger of these attacks increases in IoT networks due to limited device resources such as energy, processing power, and memory.

For this reason, modern research focuses on Physical Layer Security (PLS) which uses wireless channel characteristics to secure communication.

Traditional security systems such as Intrusion Detection Systems (**IDS**) monitor network traffic to detect suspicious activity.

However, in IoT environments these systems may be insufficient due to the distributed nature of networks.

Artificial intelligence and deep learning have recently improved IDS performance significantly

Honeypot technology is another cybersecurity technique that simulates real systems to attract attackers.

It enables researchers to monitor attacker behavior and collect valuable information about threats.

Project Objective

- The objective of this project is to develop a hybrid security system integrating IDS and Honeypot technologies
- to protect the physical layer in IoT networks against signal leakage and smart **jamming attacks**.

Project Importance

- Reviewing studies related to physical layer security in IoT networks.
- Modeling and simulating signal leakage and smart jamming attacks.
- Developing a dedicated hybrid security algorithm.
- Evaluating system performance experimentally.

Chapter One: Theoretical Study

2. Theoretical Study:

2.1 Internet of Things (IoT)

The Internet of Things refers to a network of interconnected physical devices capable of collecting and exchanging data without human intervention.

Examples include sensors, cameras, industrial systems, and smart vehicles.

Key characteristics of IoT networks:

- Large number of connected devices
- Low power consumption
- Heavy reliance on wireless communication
- Limited computational resources

These properties make traditional security solutions difficult to apply.

2.2 Physical Layer in Wireless Networks

The physical layer is responsible for transmitting signals through physical media such as air or cables.

In wireless networks, digital data is converted into electromagnetic signals using techniques such as:

- Frequency Modulation
- Phase Modulation
- Spread Spectrum

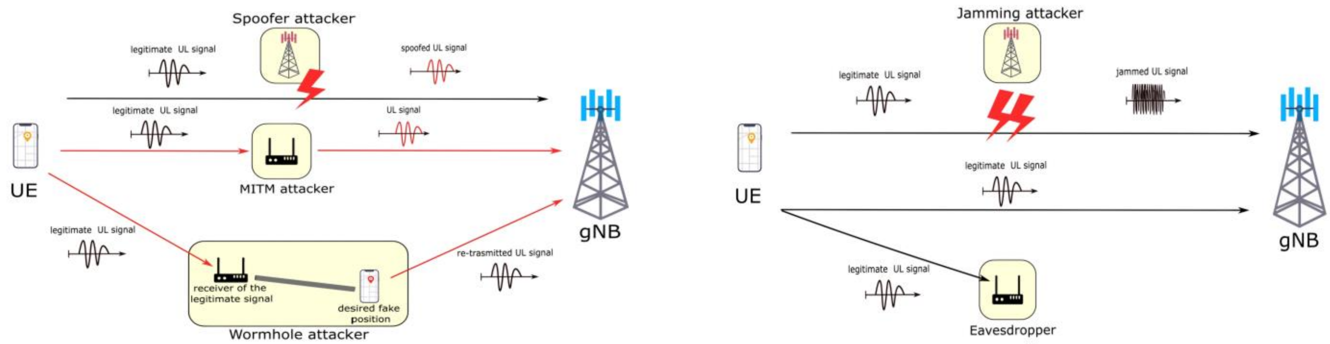
2.3 Security Threats in the Physical Layer:

Eavesdropping: attackers capture wireless signals to obtain sensitive information.

Jamming Attack: attackers transmit interference signals to disrupt communication.

Types include Constant Jamming, Reactive Jamming and Smart Jamming.

Spoofing: attackers impersonate legitimate devices.



2.4 Physical Layer Security

Physical Layer Security utilizes wireless channel characteristics to secure communications.

Techniques include Beamforming, Artificial Noise, and Reconfigurable Intelligent Surfaces (RIS).

2.5 Intrusion Detection Systems

IDS monitor systems or networks to detect cyber-attacks.

Types include Signature-Based IDS and Anomaly-Based IDS.

2.6 Honeypot Technology

A Honeypot is a decoy system designed to attract attackers and record their activities for analysis.

2.7 Hybrid IDS & Honeypot System

Combining IDS and Honeypot enables real-time attack detection and attacker behavior analysis.

Chapter Two: Previous Studies

2.8. Related Work

This chapter presents and analyses previous research and studies related to the topic of the project, which addressed the security of Internet of Things (IoT) networks, particularly at the physical layer, in addition to Intrusion Detection Systems (IDS), Honeypot technologies, and protection methods against eavesdropping and wireless jamming attacks.

The aim of this chapter is to understand the scientific developments in this field and benefit from the results of these studies in order to identify the research gaps that the current project seeks to address.

2.8.1 Computational Optimization of RIS-Enhanced Backscatter and Direct Communication for 6G IoT

Researchers:

- Syed Zain Ul Abideen
- Mian Muhammad Kamal
- Eaman Alharbi
- Ashfaq Ahmad Malik
- Wadee Alhalabi
- Muhammad Shahid Anwar
- Liaqat Ali

This study addressed the security problem in future Internet of Things networks within sixth-generation (6G) network environments, focusing on the use of Reconfigurable Intelligent Surfaces (RIS) to enhance communication performance and physical layer security.

The study proposed a computational framework that combines RIS technology with Physical Layer Security (PLS) techniques in order to improve secrecy rate and energy efficiency in wireless networks. It also used the Deep Deterministic Policy Gradient (DDPG) algorithm from reinforcement learning to optimize the configuration of intelligent surfaces and dynamically control signal direction.

The results of the study showed that integrating RIS with reinforcement learning algorithms leads to:

- Improvement in Secrecy Rate
- Increased spectral efficiency
- Reduction of interference and unwanted signals

The importance of this study lies in highlighting the role of intelligent technologies in enhancing wireless communication security in IoT networks.

2.8.2 Enhancing Physical Layer Security in WPCNs With IRS and Controlled Jamming for 6G IoT Applications

Researchers:

- Iqra Hameed
- Abid Afridi
- Myung-Sun Baek
- Hyoung-Kyu Song

This study focused on improving communication security in Wireless Powered Communication Networks (WPCNs), which are widely used in low-power IoT applications.

The study proposed a model that combines:

- Intelligent Reflecting Surfaces (IRS)
- Controlled Jamming

This approach aims to weaken the ability of eavesdroppers to receive wireless signals and improve communication secrecy.

An optimization algorithm based on Alternating Optimization techniques was also developed to adjust:

- Energy and information transmission time
- Phase coefficients in IRS elements
- Jamming power directed toward eavesdroppers

Enhancing_Physical_Layer_Securi…

The results showed that combining IRS with controlled jamming significantly improves secrecy rate and energy efficiency in IoT networks.

2.8.3. LoRaWAN Physical Layer-Based Attacks and Countermeasures

Researchers:

- Henri Ruotsalainen
- Guanxiong Shen
- Junqing Zhang
- Radek Fujdiak

This study presented a comprehensive review of attacks targeting the physical layer in LoRaWAN networks, which are among the most widely used communication protocols in IoT.

The study analyzed several types of attacks including:

- Wireless Jamming attacks
- Battery exhaustion attacks
- Data manipulation attacks
- Replay attacks

It also discussed several protection mechanisms such as:

- Device fingerprinting-based authentication
- Key generation using wireless channel characteristics

- Wireless jamming detection systems

LoRaWAN_Physical_Layer-Based_At...

This study confirms that the physical layer represents an important vulnerability in IoT networks, which requires the development of advanced security techniques to address these attacks.

2.8.4 An Innovative Honeypot Architecture for Detecting and Mitigating Hardware Trojans in IoT Devices

Researchers:

- Amira Hossam Eldin Omar
- Hassan Soubra
- Donatien Koulla Moulla
- Alain Abran

This study investigated the use of Honeypot technology to detect security threats in IoT devices, particularly attacks related to hardware and electronic circuits such as Hardware Trojans.

The study proposed a Honeypot system architecture that simulates real devices in order to attract attackers and analyze their behavior without affecting real systems.

The system was implemented using:

- Raspberry Pi device
- FPGA platform for attack simulation

The results showed that using honeypots helps to:

- Detect threats early
- Analyze attacker behavior
- Improve protection strategies in IoT networks.

2.8.5 A Review of Physical Layer Security in Aerial–Terrestrial Integrated Internet of Things

Researchers:

- Yixin He
- Jingwen Wu
- Lijun Zhu
- Fanghui Huang
- Baolei Wang
- Deshan Yang
- Dawei Wang

This study examined physical layer security in IoT networks that integrate terrestrial systems with aerial platforms such as drones.

The study discussed security threats faced by these networks such as:

- Eavesdropping on communications
- Wireless interference
- Spoofing
- Denial-of-Service attacks

It also presented several proposed techniques to enhance security such as:

- Cooperative communications
- Covert communications
- Intelligent reflecting surfaces
- Beamforming techniques

drones-09-00312-v2

2.8.6 Wireless PHY-Layer Threats to Security and Privacy for IoT Systems

Researchers:

- Renato Lo Cigno

- Francesco Gringoli
- Stefania Bartoletti
- Marco Cominelli
- Lorenzo Ghiro
- Samuele Zanini

This study examined security threats at the physical layer associated with Integrated Sensing and Communication (ISAC) technologies.

The study explained that wireless signals can be exploited not only for data transmission but also for extracting information about the environment or users, which may lead to privacy violations.

It also discussed several types of attacks such as:

- Inferring user information from signals
- Manipulating location information
- Signal analysis-based attacks

2.8.7 A Physics-Based Hyper Parameter Optimized Federated Multi-Layered Deep Learning Model for Intrusion Detection in IoT Network

Researchers:

- Chirag Jitendra Chandnani
- Vedik Agarwal
- Shlok Chetan Kulkarni
- Aditya Aren
- D. Geraldine Bessie Amali
- Kathiravan Srinivasan

This study proposed an advanced Intrusion Detection model for IoT networks using Deep Learning and Federated Learning.

The proposed system trains intrusion detection models in a distributed manner across devices without sending sensitive data to a central server.

The model was tested using several cybersecurity attack datasets, and the results showed that the system achieved an accuracy of up to 99% in detecting attacks.

A_Physics-Based_Hyper_Parameter…

This study highlights the importance of using artificial intelligence techniques to improve IDS performance in IoT environments.

2.8.8 Analysis of Previous Studies

Through reviewing the previous studies, it can be observed that most research focused either on enhancing physical layer security or developing intrusion detection systems separately.

Some studies focused on advanced techniques such as:

- RIS intelligent surfaces
- Directed jamming
- Beamforming techniques

While other studies focused on developing intrusion detection systems using artificial intelligence and deep learning.

Some studies also highlighted the importance of Honeypot technology for attracting attackers and analyzing their behavior in order to improve defense strategies.

However, most of these studies addressed these technologies separately, and they were not integrated into a comprehensive security system capable of detecting and analyzing attacks simultaneously.

2.8.9 Justification of the Current Project

Based on the results of previous studies, several points highlight the need for the proposed project:

1. Most studies focused on protecting the physical layer using specific techniques without integrating them with intrusion detection systems.

2. Some research used IDS or artificial intelligence without focusing on physical layer attacks such as jamming and eavesdropping.
3. The use of Honeypot technology in IoT networks is still limited in the field of physical layer protection.

Accordingly, the current project aims to develop a hybrid security system combining IDS and Honeypot technologies to protect the physical layer in IoT networks against signal leakage and smart jamming attacks, thereby helping to fill the research gap that previous studies have not fully addressed.

How to benefit from it in the project	Features	What has been accomplished	Journal	Date	Author	The Article
We use this paper to define the physical layer threat models our project addresses, especially passive sensing attacks undetectable by conventional means. Its attack classification in ISAC networks forms the threat base for our IDS, and its identified signal characteristics serve as detection features within our Honeypot	Recently considered a key reference focusing on passive sensing as a hidden threat undetectable by traditional means, offering defensive strategies based on signal characteristics to secure IoT devices	A comprehensive analysis of physical layer vulnerabilities arising from the integration of communications with wireless sensing (ISAC) along with classification of modern attacks such as eavesdropping and unauthorized sensing in next-generation networks	MDPI	2025	Renato Lo Cigno Francesco Gringoli Et al	Communication and Sensing: Wireless PHY-Layer Threats to Security and Privacy for IoT Systems and Possible Countermeasures
We adopt its comprehensive PLS framework to design our defensive mechanisms, and use its anti-eavesdropping methods to enhance the Honeypot's ability to simulate realistic signal environments that expose attackers. Its power distribution techniques also inform our experimental setup	Integrates AI techniques and Reconfigurable Intelligent Surfaces (RIS) to secure wireless communications, and highlights the importance of optimizing flight paths to enhance data security against attackers in open environments	A comprehensive review of physical layer security techniques in hybrid networks connecting unmanned aerial vehicles with ground-based devices, covering anti-eavesdropping and anti-jamming methods through optimized antenna design and power distribution.	MDPI	2025	Yixin He Et al	A Review of Physical Layer Security in Aerial-Terrestrial Integrated Internet of Things
We directly adapt its multi-layered deep learning architecture for detecting jamming and signal leakage in our IDS. Its federated learning approach preserves data privacy across distributed IoT devices, and its physics-based optimization algorithm tunes hyperparameters to suit resource-constrained devices	Combines high detection accuracy with data privacy preservation, outperforming traditional models in training speed and resource consumption, making it ideal for resource-constrained IoT devices	Development of an Intrusion Detection System (IDS) operating within a federated learning environment to ensure data privacy, using a multi-layered deep learning model with a physics-based optimization algorithm for hyperparameter tuning, thereby increasing attack detection accuracy in distributed systems	MDPI	2025	Chirag Jitendra Chandnani Et al	A Physics-Based Hyper Parameter Optimized Federated Multi-Layered Deep Learning Model for Intrusion Detection in IoT Networks

This is our primary design reference for the Honeypot component We build upon its device behavior monitoring and activity isolation infrastructure, adopt its damage mitigation mechanism, and extend its active defense concept to collect attacker intelligence and isolate compromised devices	One of the few papers linking the Honeypot concept with IoT hardware security, providing an active defensive strategy that protects the network by isolating infected devices and collecting data on the attacker's methodology	Design of an advanced Honeypot infrastructure capable of detecting internal threats such as Hardware Trojans embedded in device hardware. The system relies on monitoring device behavior, isolating suspicious activity, and providing a mitigation mechanism to minimize attack damage	MDPI	2024	Amira Hossam Eldin Omar ET AL	An Innovative Honeypot Architecture for Detecting and Mitigating Hardware Trojans in IoT Devices
We use its documented attack scenarios — selective jamming and packet blocking via SDR — as realistic test cases within our Honeypot environment. Its analysis of LoRaWAN vulnerabilities directly shapes our system's evaluation framework	Presents realistic attack scenarios that can be simulated in the project, and proposes solutions to increase network resilience against intentional disruption attempts	A practical analysis of vulnerabilities in the LoRaWAN protocol against physical layer attacks such as selective jamming and packet blocking, demonstrating how an attacker using SDR can disable thousands of IoT devices	MDPI	2022	Luis Barbira ET AL	LoRaWAN Physical Layer-Based Attacks: Vulnerabilities and Countermeasures
We integrate the RIS concept as a defensive layer to steer signals away from attackers, and adopt the DDPG algorithm to dynamically optimize our hybrid system's decisions based on real-time radio environment changes, while applying its energy-saving mechanisms for resource-constrained IoT devices	Provides a smart and dynamic solution that adapts to rapid changes in the radio environment, making it a strong reference for using deep reinforcement learning in threat detection and securing communications while balancing energy efficiency with maximum data confidentiality — a fundamental requirement for resource-limited IoT devices	Development of a model to enhance the security and efficiency of communications in 6G IoT networks. Researchers used the Deep Deterministic Policy Gradient (DDPG) algorithm to optimize the operation of Reconfigurable Intelligent Surfaces (RIS), ensuring efficient signal routing to legitimate devices while protecting them from physical layer eavesdropping, alongside integrating direct and backscatter communication techniques to reduce energy consumption	CMES	2025	Syed Zain Ul Abideen ET AL	Computational Optimization of RIS-Enhanced Backscatter and Direct Communication for 6G IoT: A DDPG-Based Approach with Physical Layer Security

We adopt its Secrecy Rate mathematical formulation as a performance metric for our system. Its Controlled Jamming concept is integrated into our Honeypot to degrade the attacker's data collection ability, and its IRS strategy strengthens our system's ability to distinguish legitimate users from attackers	Introduces the concept of defensive jamming, where jamming is used as a tool to protect privacy rather than disrupt the network. The paper provides precise mathematical equations for maximizing the Secrecy Rate, which benefits the project when designing the Honeypot mechanism aimed at luring the attacker and weakening its ability to collect real data	Proposal of an advanced defensive strategy to protect Wireless Powered Communication Networks (WPCNs) from eavesdroppers, relying on Intelligent Reflecting Surfaces (IRS) to strengthen the legitimate user's signal while simultaneously transmitting Controlled Jamming signals to confuse any eavesdropping device, with optimized power distribution to ensure device sustainability	IEEE Access	2025	Iqra Hameed Et al	Enhancing Physical Layer Security in WPCNs With IRS and Controlled Jamming for 6G IoT Applications

1.reference studies table

Chapter Three: The Proposed Solution

3. Proposed Solution

3.1 System Overview

The proposed system aims to enhance the security of Internet of Things (IoT) networks at the physical layer by developing a hybrid security framework that integrates both an Intrusion Detection System (IDS) and a Honeypot environment.

The system is implemented on the Ubuntu operating system, which provides a stable and flexible environment for integrating multiple tools and technologies. The main objective of this system is to detect malicious activities such as signal leakage and smart jamming attacks, and then redirect attackers toward a controlled honeypot environment for further analysis.

To achieve seamless integration between all components, the Python programming language is used as the core development tool. Python enables communication and coordination between different modules, including simulation tools, detection models, and the honeypot system.

3.2 System Architecture

The proposed architecture consists of four main components:

1. IoT Network Simulation Layer

2. Physical Layer Simulation

3. Intrusion Detection System (IDS)

4. Honeypot Environment

These components are integrated together to form a hybrid security system capable of detecting and analyzing cyber-attacks in IoT networks.

3.3 Tools and Technologies Used

Several tools and technologies were used to build and implement the proposed system:

3.3.1 Ubuntu Operating System

Ubuntu is used as the main operating system for the project due to its compatibility with networking and simulation tools, as well as its support for open-source security frameworks.

3.3.2 Python Programming Language

Python is used as the primary programming language to integrate all system components. It is responsible for:

- Connecting different modules (NS-3, GNU Radio, IDS, and Honeypot)
- Processing data and managing workflows
- Automating system operations

3.3.3 NS-3 with NetAnim

NS-3 is used as an IoT network simulator, while NetAnim is used for visualization. These tools are utilized for:

- Building the IoT network topology
- Simulating communication between nodes
- Generating network traffic data (PCAP files)

3.3.4 GNU Radio

GNU Radio is used to simulate the physical layer of the communication system. It is responsible for:

- Generating wireless signals
- Adding noise to the communication channel
- Simulating physical layer attacks such as:
- Signal leakage
- Smart jamming

3.3.5 Intrusion Detection System (IDS)

The IDS is designed to analyze both signal-level and network-level data to detect malicious activities. It is trained using extracted features from simulated attack scenarios.

3.3.6 Honeypot Environment

The honeypot is used as a deceptive security mechanism that:

Simulates fake IoT devices

Attracts attackers after detection by the IDS

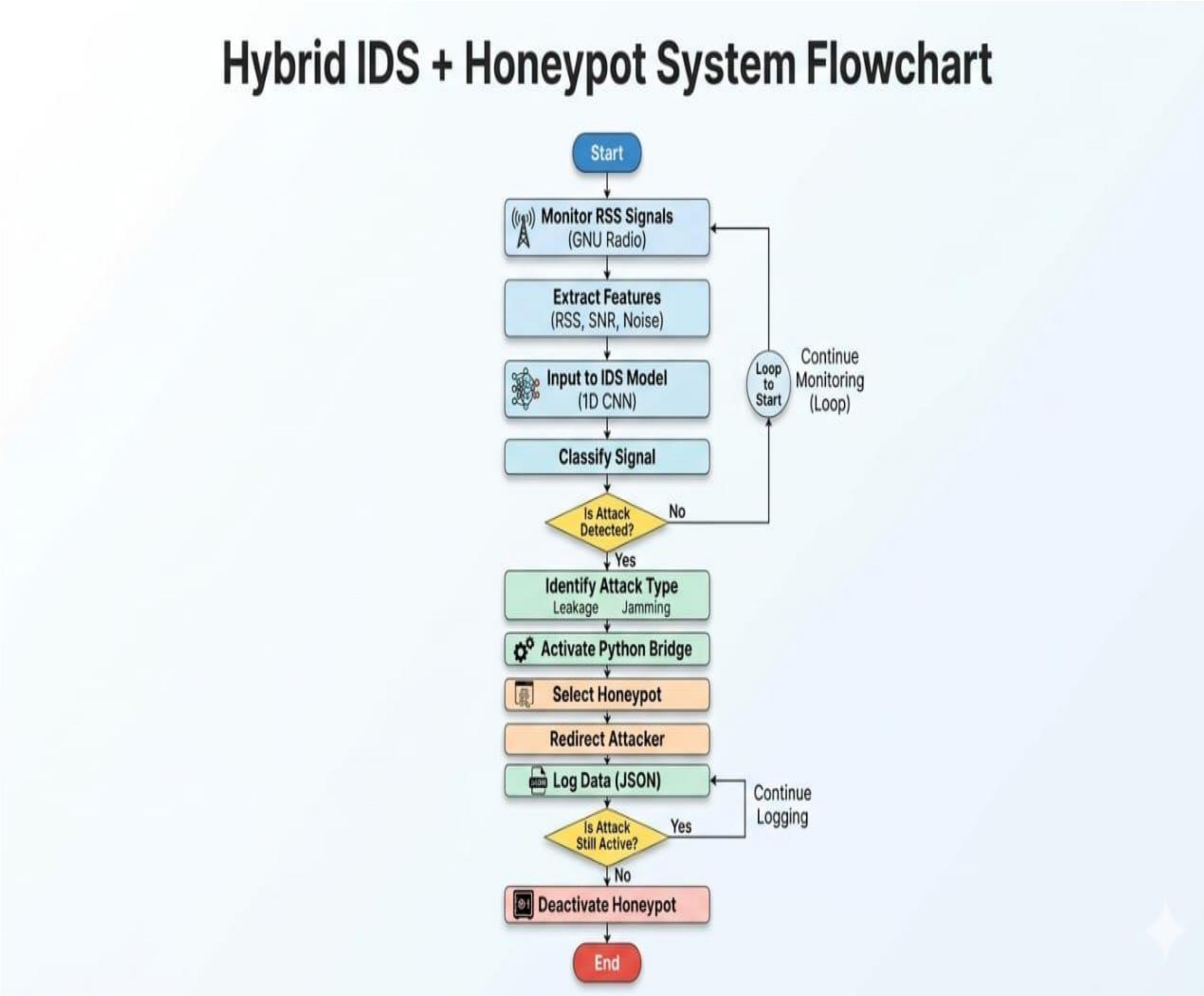
Logs and analyzes attacker behavior automatically

3.4 System Workflow

The proposed system operates through the following sequence:

1. The IoT network topology is created using NS-3.
2. Wireless signals are generated and manipulated using GNU Radio.
3. Attack scenarios such as signal leakage and smart jamming are simulated.
4. Network and signal data are collected and processed.
5. The IDS analyzes the data and detects potential attacks.
6. Once an attack is detected, the attacker is redirected to the honeypot environment.
7. The honeypot records attacker behavior for further analysis.

(Flowchart Representation)



3.5 Integration Strategy

All components are integrated into a single hybrid system using Python scripts. The integration ensures:

- Real-time communication between modules
- Efficient data exchange between simulation and detection systems
- Automated response to detected threats

- The final system provides a comprehensive solution for detecting, analyzing, and responding to physical layer attacks in IoT networks.

3.6 System Diagram

The following figure illustrates the tools used and the overall architecture of the proposed system:

3.6.1Tools Used

Ubuntu: Operating system	Python: Main integration language
GNU Radio: - Wireless signals - Noise - Attack simulation	NS-3: - Topology - Traffic - PCAP
Tiny Ml: analyzes rss	Cowrie: Honeypot environment

2.Tools Used

3.6.2Project Implementation Plan

Build IoT topology	Generate signals (GNU Radio)
Create attack scenarios + features	Train IDS + Honeypot
Integrate full system	Evaluate results

3. **Project Implementation Plan**

Chapter Four: Practical Application

Chapter 4:

4.Practical Implementation

4.0 Physical Layer Signal Analysis (RSS & SNR)

4.0.1 Overview of Physical Layer Analysis

This section presents a detailed analysis of the physical layer characteristics used in the proposed system. The analysis focuses on comparing different communication scenarios, including Normal operation, Signal Leakage, and Smart Jamming attacks, using key signal metrics such as Received Signal Strength (RSS) and Signal-to-Noise Ratio (SNR).

The results are obtained from GNU Radio simulations and demonstrate clear distinctions between normal and malicious signal behaviors.

4.0.2 Received Signal Strength (RSS)

Definition

Received Signal Strength (RSS) represents the power level of a received wireless signal at the receiver. It is measured in decibels relative to one milliwatt (dBm).

Physical Interpretation

RSS values provide an indication of signal quality:

RSS Value (dBm)	Interpretation
-30 dBm	Excellent signal
-50 dBm	Very strong signal
-65 dBm	Good signal (typical for smart homes)
-75 dBm	Weak but acceptable
-90 dBm	Very weak (connection may be lost)

RSS Calculation

RSS is calculated using the following equation:

RSS = 10 × log10 (Signal Power / Reference Power)

4.0.3 RSS Comparison (Wi-Fi 2.4 GHz)

The following results illustrate RSS behavior under different scenarios:

Scenario	Average RSS	Description
Normal	-65 dBm	Stable signal with low fluctuations
Signal Leakage	-67 dBm	Slight degradation due to leakage
Smart Jamming	-80 dBm	Significant drop due to interference

Observation:
Smart jamming introduces sudden drops in RSS values, often occurring intermittently after a short period (e.g., 10 seconds), making detection more challenging.

4.0.4 RSS Comparison (LoRa 868 MHz)

LoRa signals exhibit different behavior due to their physical characteristics:

Scenario	Average RSS	Description
Normal	-70 dBm	Naturally weaker than Wi-Fi
Signal Leakage	-72 dBm	Slight degradation
Smart Jamming	-78 dBm	Moderate interference

4.0.5 Signal-to-Noise Ratio (SNR)

Definition

Signal-to-Noise Ratio (SNR) is the ratio between signal strength and background noise, measured in decibels (dB).

Interpretation

Scenario	SNR Value	Interpretation
Normal	25 dB	Clear and high-quality signal
Signal Leakage	23 dB	Slight degradation
Smart Jamming	10 dB	Highly noisy signal

Relationship with RSS

SNR is related to RSS as follows:

SNR = RSS_signal − RSS_noise

4.0.6 Performance Comparison Between Scenarios

A comparative analysis of all scenarios is presented below:

Metric	Normal	Signal Leakage	Smart Jamming
RSS (Wi-Fi)	-65 dBm	-67 dBm	-80 dBm
RSS (LoRa)	-70 dBm	-72 dBm	-78 dBm
SNR	25 dB	23 dB	10 dB
RSS Variance	Low	Medium	High
Detection Difficulty	Low	Medium	High

4.0.7 Causes of Signal Leakage in IoT Devices

Signal leakage occurs due to several hardware-related factors:

- Poor circuit design
- Inefficient filters
- Harmonic interference
- Weak isolation between components

These issues result in unintended electromagnetic emissions that can be exploited by attackers.

4.0.8 Smart Jamming Attack

Definition

Smart Jamming is an adaptive and intelligent attack that dynamically interferes with wireless communication.

Characteristics

Feature	Traditional Jamming	Smart Jamming
Pattern	Random / Constant	Adaptive
Frequency Usage	Full spectrum	Targeted
Timing	Continuous	Pulsed / Intermittent
Detection	Easy	Difficult
Impact	General disruption	Targeted disruption

4.0.9 Scenario Implementation in the Project

In the implemented simulation:

- The attacker (Node 9 in NS-3) starts the attack at time t = 10 seconds
- Transmission frequency: 2.412 GHz (Wi-Fi)
- Attack pattern: Pulsed (every 0.05 seconds)
- Packet size: 1024 bytes

This setup simulates realistic attack behavior in IoT environments.

4.0.10 General Observation

The results confirm that:

- RSS and SNR are effective indicators for detecting physical layer attacks
- Smart Jamming is significantly more difficult to detect than traditional interference

- Combining multiple features enhances detection accuracy

4.0.11 RSS Distribution per Scenario

To further analyze signal behavior, the distribution of RSS values was examined across different scenarios using a histogram-based representation. This analysis provides deeper insight into signal stability and variability under Normal conditions, Signal Leakage, and Smart Jamming attacks.

Distribution Characteristics

Scenario	Center (Mean RSS)	Spread (Variance)	Interpretation
Normal	-65 dBm	Narrow (2–4 dB)	Stable and consistent signal
Signal Leakage	-67 dBm	Medium (4–6 dB)	Slight signal degradation
Smart Jamming	-75 dBm	Wide (8–12 dB)	Highly unstable signal

Analysis of Results

- Normal Scenario:
  The RSS distribution is narrow and centered around a stable value, indicating a consistent and reliable signal with minimal fluctuations.
- Signal Leakage:
  A slight shift toward lower RSS values is observed, along with moderate spreading, reflecting minor signal degradation due to leakage effects.
- Smart Jamming:
  The distribution becomes significantly wider and shifts to lower RSS values, indicating strong interference and high variability, which is a key indicator of an active attack.

4.1 Phase One: Simulation Environment Setup

4.1.1 Smart Home IoT Network Topology

A hybrid smart home IoT network was designed using the NS-3 network simulator, combining both Wi-Fi and LoRa communication protocols. The network consists of ten nodes distributed as follows:

Wi-Fi Nodes (IEEE 802.11b, 2.4 GHz):

- Central Gateway (Node 0): Acts as the main Access Point
- Camera (Node 1): Sends data every second (512 bytes)
- Thermostat (Node 2): Sends data every 5 seconds
- Smart Door (Node 3): Sends data every 3 seconds
- Fire Sensor (Node 4): Sends data every 2 seconds
- LoRa Gateway (Node 5): Collects data from LoRa devices and forwards it to the main gateway

LoRa Nodes (868 MHz):

- LoRa-Garden Sensor (Node 6)
- LoRa-Electric Meter (Node 7)
- LoRa-OutDoor Sensor (Node 8)

Attacker Node:

- Attacker (Node 9): Initiates the attack after 10 seconds from the start of the simulation

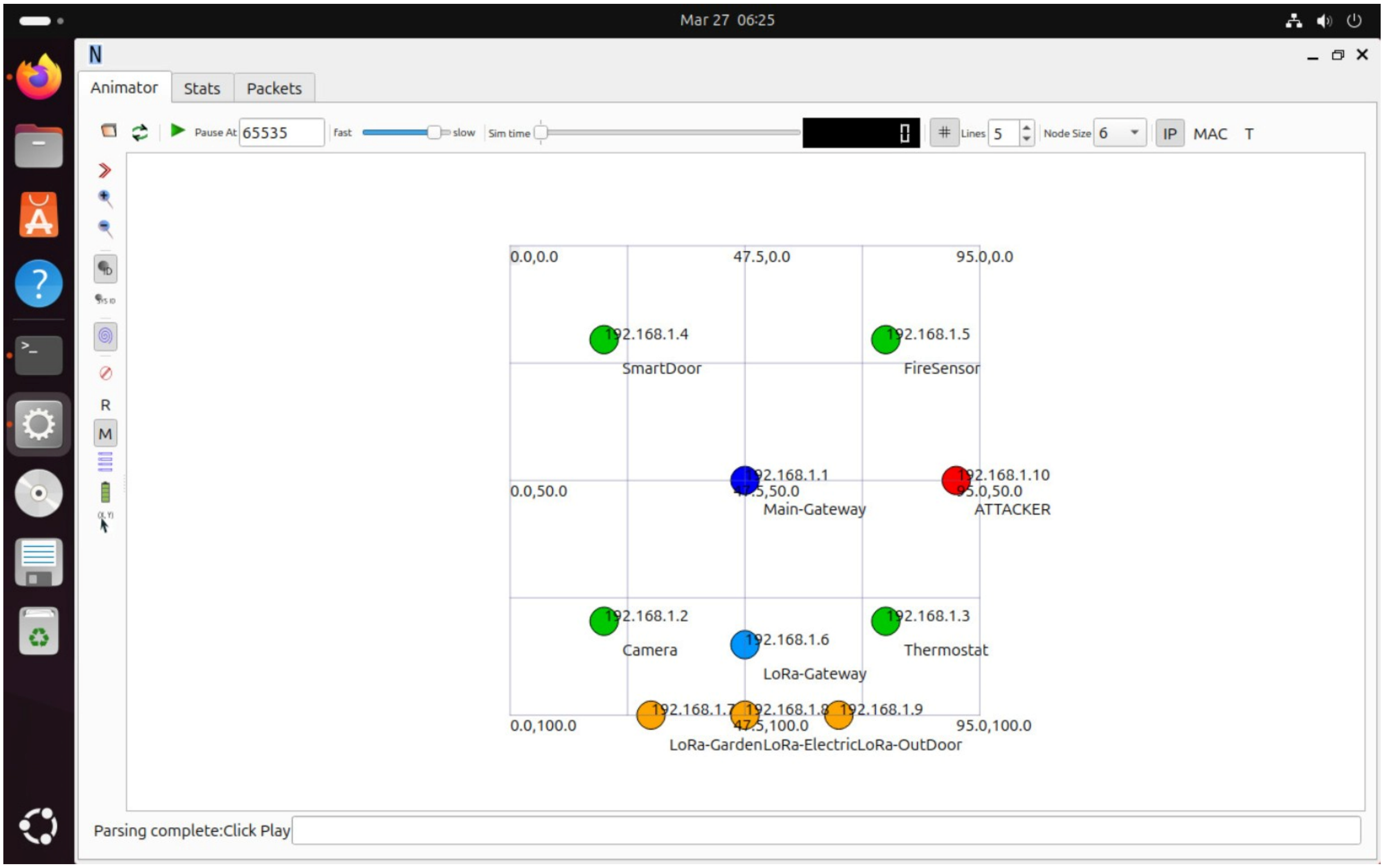


Figure 4.1:

Smart Home IoT Network Topology (NS-3 NetAnim Visualization)

4.1.2 NS-3 Simulation Parameters

The simulation environment was configured with the following parameters:

Parameter	Value
Simulation Time	40 seconds
Wi-Fi Protocol	IEEE 802.11b
Data Rate	1 Mbps
Network Addressing	192.168.1.0/24
Transport Protocol	UDP
Port	9

(4)

The nodes were positioned in a 2D spatial layout using NetAnim, where the gateway is located at the center and other devices are distributed around it to simulate a realistic smart home environment.

4.2 Phase Two: Physical Layer Simulation

4.2.1 Wireless Signal Generation using GNU Radio

GNU Radio (version 3.10.9) was used to simulate the physical layer behavior of the IoT network. Six simulation models were developed to represent different communication scenarios for both Wi-Fi and LoRa.

Wi-Fi Signals (2.4 GHz)

A sinusoidal signal with a frequency of 2400 Hz was generated to represent Wi-Fi transmission. The signal was passed through a channel_model block with different parameters depending on the scenario:

Scenario	noise_voltage	frequency_offset	epsilon
Normal	0.002	0.0	1.0
Signal Leakage	0.008	0.0002	1.0
Smart Jamming	0.040	0.001	1.002

(5)

LoRa Signals (868 MHz)

A sinusoidal signal with a frequency of 868 Hz and lower amplitude (0.03 compared to 0.05 for Wi-Fi) was used to reflect LoRa characteristics:

Scenario	noise_voltage	frequency_offset	epsilon
Normal	0.003	0.0	1.0
Signal Leakage	0.010	0.0001	1.0

Smart Jamming	0.050	0.0015	1.003

(6)

4.2.2 Realistic RSS Extraction

From each window of 100 samples, key physical layer metrics were extracted.

Received Signal Strength (RSS)

RSS values were calculated based on the average signal power and converted to **dBm**, while applying realistic constraints:

Protocol	Scenario	RSS Range
Wi-Fi	Normal	-45 to -60 dBm
Wi-Fi	Signal Leakage	-60 to -72 dBm
Wi-Fi	Smart Jamming	-72 to -90 dBm
LoRa	Normal	-80 to -100 dBm
LoRa	Signal Leakage	-100 to -115 dBm
LoRa	Smart Jamming	-115 to -130 dBm

(7)

These ranges are consistent with real-world IoT wireless communication standards.

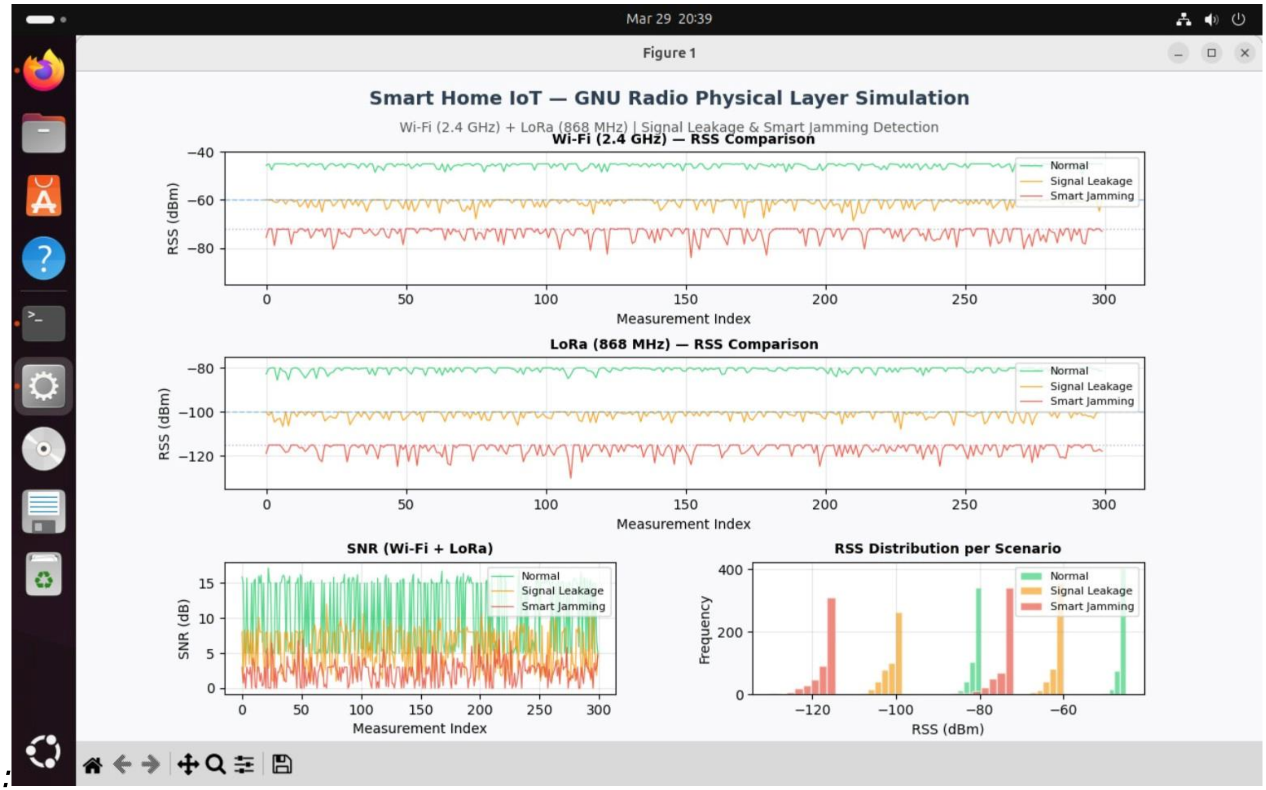


Figure 4.2

RSS and SNR Analysis for Wi-Fi and LoRa under Different Scenarios

4.3 Phase Three: Dataset Construction

4.3.1 Multi-Layer Feature Fusion Approach

A multi-layer feature fusion approach was adopted, where features from different layers were extracted and combined into a single dataset.

Physical Layer Features (GNU Radio):

Feature	Description	Unit
rss_dbm	Received Signal Strength	dBm
snr_db	Signal-to-Noise Ratio	dB
noise_floor	Background noise level	dBm
freq_offset	Frequency offset	rad
protocol	Protocol type (Wi-Fi=0, LoRa=1)	-

(8)

Network Layer Features (NS-3):

Feature	Description	Unit
packet_loss_rate	Packet loss rate	%
avg_delay_ms	Average delay	ms
retry_count	Number of retransmissions	count
avg_packet_size	Average packet size	bytes
std_packet_size	Standard deviation of packet size	bytes
avg_inter_arrival	Average inter-arrival time	sec

(9)

4.3.2 Final Dataset Characteristics

Property	Value
Total Records	3000
Number of Features	12
Label Distribution	Balanced (1000 per class)
File Size	~350 KB
Format	CSV

(10)

Class Distribution:

- Label 0 — Normal (33.3%)
- Label 1 — Signal Leakage (33.3%)
- Label 2 — Smart Jamming (33.3%)

This perfect balance eliminates the class imbalance problem, improving machine learning performance.

4.4 Phase Four: Current Results

4.4.1 NS-3 Simulation Results

The smart home simulation produced the following results:

- Packet loss rate under normal conditions: 0%
- Average delay: between 3 ms and 42 ms, depending on the device
- The attacker node generates high traffic:
  - Packet rate: every 0.1 seconds
  - Packet size: 1024 bytes

4.4.2 GNU Radio Results

A clear separation between the three scenarios (Normal, Signal Leakage, Smart Jamming) is observed in the RSS plots.

This confirms that physical layer features are highly effective in distinguishing attack scenarios, which is essential for training the IDS model.

4.6 Phase Five: Dataset Integration and Model Preparation

4.6.1 Dataset Sources

In this phase, two types of datasets were used to enhance the performance and reliability of the Intrusion Detection System (IDS):

1. **Pre-existing Dataset:**
   A ready-made dataset was obtained from a previous research study (file: *3567445.3567456.pdf*). This dataset contains labeled network traffic data representing different attack scenarios and normal behavior.
2. **Generated Dataset (Proposed System):**
   A custom dataset was generated using the proposed simulation environment (NS-3 + GNU Radio). This dataset includes both:
   - **Physical layer features** (RSS, SNR, noise, frequency offset)
   - **Network layer features** (packet loss, delay, retransmissions, etc.)

The combination of both datasets improves the generalization capability of the IDS model.

4.6.2 Dataset Generation Using GNU Radio

Wireless signal data was generated using GNU Radio under different channel conditions, including:

- Normal communication
- Signal leakage scenario
- Smart jamming attacks

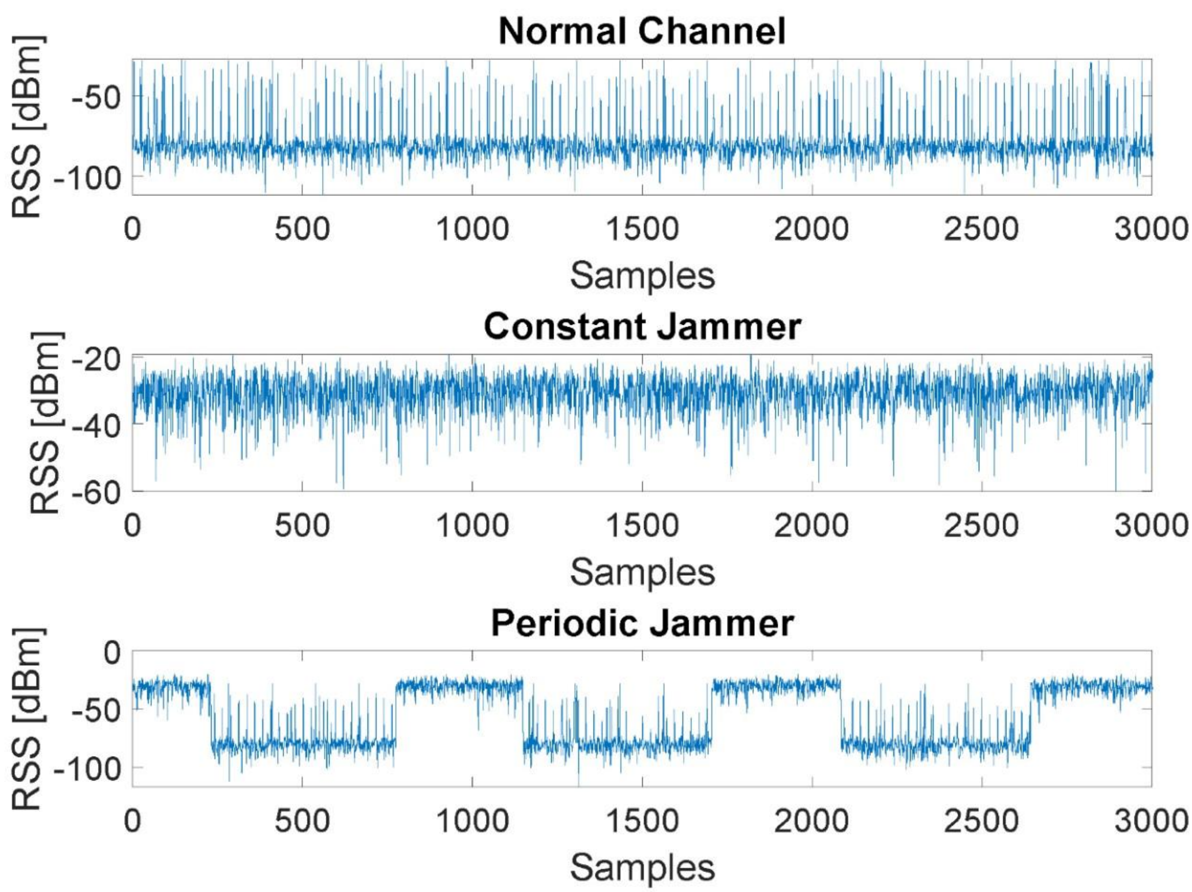


Figure 4.4:

GNU Radio Flowgraph for Wireless Signal Acquisition and Processing

This flowgraph shows how signals are captured, processed using FFT blocks, and stored into files for further analysis.

4.6.3 Signal Behavior under Different Scenarios

The generated dataset includes RSS measurements under different channel conditions. The following figure illustrates the differences between normal operation and jamming attacks:

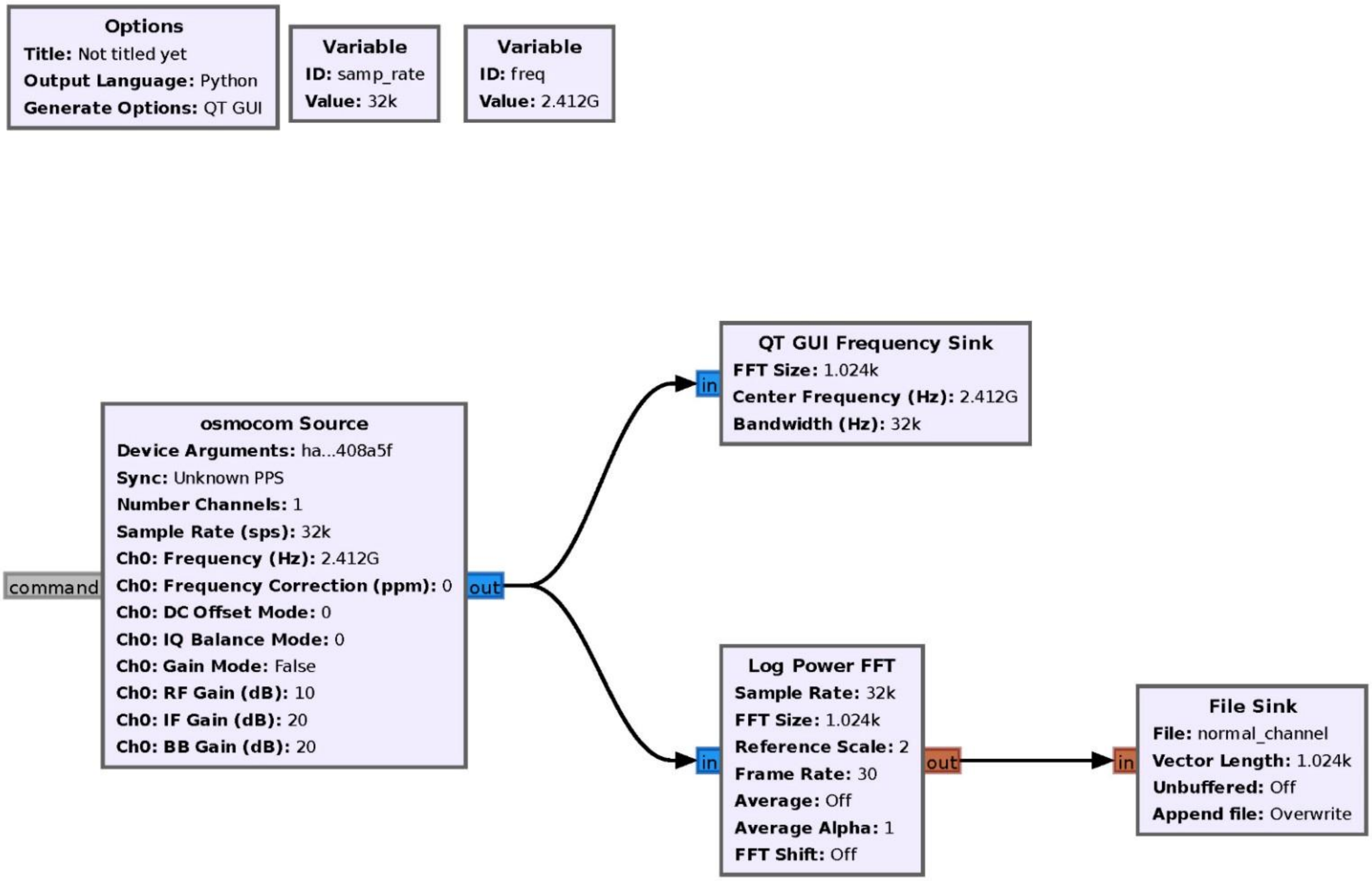


Figure 4.5:

RSS Comparison for Normal Channel, Constant Jammer, and Periodic Jammer

Observations:

- **Normal Channel:**
  RSS values remain relatively stable with small fluctuations around a baseline level.
- **Constant Jammer:**
  RSS values significantly increase and become noisy due to continuous interference.
- **Periodic Jammer:**
  RSS values alternate between normal and jammed states, reflecting periodic attack behavior.

These differences confirm that **physical layer features are effective indicators for detecting attacks**.

4.6.4 Dataset Combination and Preprocessing

The final dataset used for training was constructed by combining:

- Features extracted from the generated simulation data
- Features from the pre-existing dataset

Preprocessing Steps:

- Data normalization
- Label encoding (Normal, Signal Leakage, Smart Jamming)
- Feature alignment between datasets
- Removal of outliers and noise

This ensures that the dataset is clean, balanced, and suitable for machine learning.

4.6.5 Model Training Strategy

The IDS model was trained using:

- The generated dataset (from GNU Radio + NS-3)
- The external dataset from previous research

This hybrid training approach provides:

- Better detection accuracy
- Improved robustness against unseen attacks
- Enhanced generalization across different scenarios

4.6.6 Importance of Dataset Integration

Integrating both simulated and real-world datasets provides several advantages:

- Captures realistic attack behavior
- Improves model performance
- Reduces overfitting
- Enhances adaptability to real IoT environments

4.7 IDS Model Training

4.7.1 Model Overview

A 1D Convolutional Neural Network (1D-CNN) was designed and trained to detect wireless jamming attacks at the physical layer in IoT networks.

This model was selected for two main reasons:

- Its proven ability to process sequential time-series data such as RSS values
- Its lightweight architecture, making it suitable for deployment on resource-constrained IoT devices within the TinyML framework

4.7.2 Dataset

4.7.2.1 Source and Description

A real-world dataset was used, collected using GNU Radio combined with Software Defined Radio (SDR) hardware. The dataset was published in a reference research paper presented at the ACM IoT Conference (2022).

The dataset consists of three text files containing continuous RSS measurements in dBm:

File	Scenario	Samples	Mean RSS	Std Dev	Label
normal_channel.txt	Normal	410,000	−80.48 dBm	12.63	0
constant_jammer.txt	Constant Jamming	410,000	−30.73 dBm	5.55	1
periodic_jammer.txt	Smart/Periodic Jamming	410,000	−59.96 dBm	25.54	2

4.7.2.2 Class Separation

The statistical characteristics clearly distinguish the three classes:

- **Normal Channel:** Low RSS (~−80 dBm) with moderate variance

- **Constant Jamming:** High RSS (~−30 dBm) due to continuous interference
- **Smart Jamming:** High variance (Std = 25.54) due to intermittent attack patterns

4.7.2.3 Data Preparation

- 80,000 samples were selected from each class
- Balanced dataset (equal distribution)
- RSS values clipped to range: **[−105, −20] dBm**
- Outliers removed

This ensures a clean and unbiased dataset for training.

4.7.3 Feature Extraction — Sliding Window

4.7.3.1 Concept

Instead of using individual RSS values, a sliding window approach was used:

- Each input contains 16 consecutive samples
- Captures temporal patterns
- Essential for detecting smart jamming behavior

Parameters:

Parameter	Value	Purpose
Window Size	16	Capture short-term dynamics
Step Size	8	50% overlap
Input Shape	(16,1)	Time-series format

4.7.4 Overfitting Prevention Strategy

4.7.4.1 Gaussian Noise Injection

- Noise added with standard deviation = 8 dBm
- Simulates real-world wireless noise

- Improves robustness

4.7.4.2 Mixup Augmentation

- Applied to 40% of samples
- Mixing ratio: 30%
- Improves decision boundaries

4.7.4.3 Network Regularization

- Dropout (0.50 – 0.55)
- L2 Regularization (0.08)
- ReduceLROnPlateau

These techniques reduce overfitting and improve generalization.

4.7.5 Model Architecture

The proposed 1D-CNN model consists of:

Layer	Type	Parameters	Purpose
Input	—	(16,1)	RSS window
Conv1D	CNN	8 filters	Feature extraction
BatchNorm + Pooling	—	—	Stabilization
Conv1D	CNN	16 filters	High-level features
GlobalAvgPool	—	—	Feature compression
Dense	Fully Connected	12 units	Classification
Output	Softmax	3 units	Class prediction

4.7.6 Training Procedure

4.7.6.1 Data Splitting

Set	Ratio	Purpose

Training	70%	Model learning
Validation	15%	Hyperparameter tuning
Test	15%	Final evaluation

4.7.6.2 Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	0.0003
Loss Function	Sparse Categorical Crossentropy
Batch Size	256
Epochs	40
Early Stopping	8 epochs

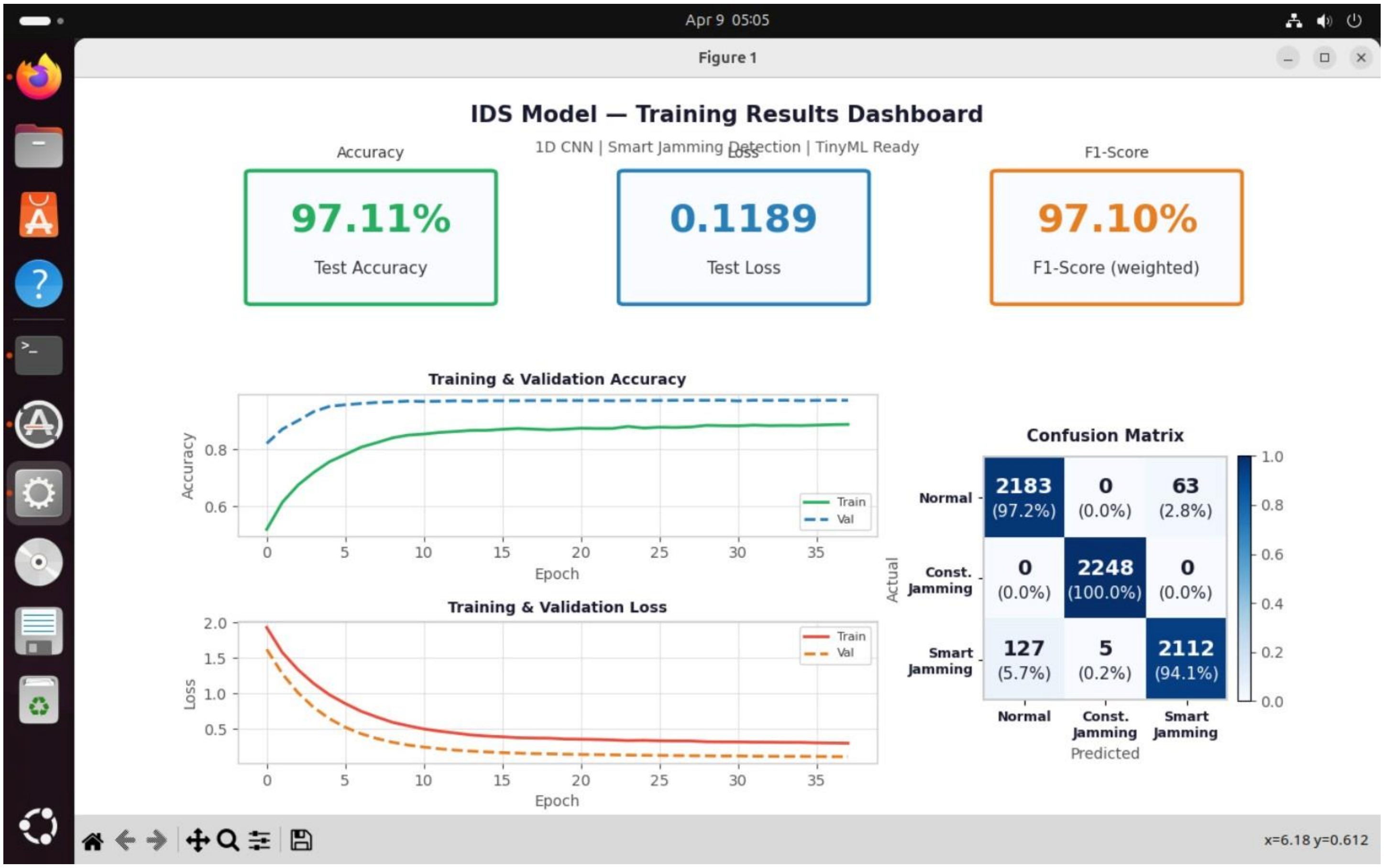


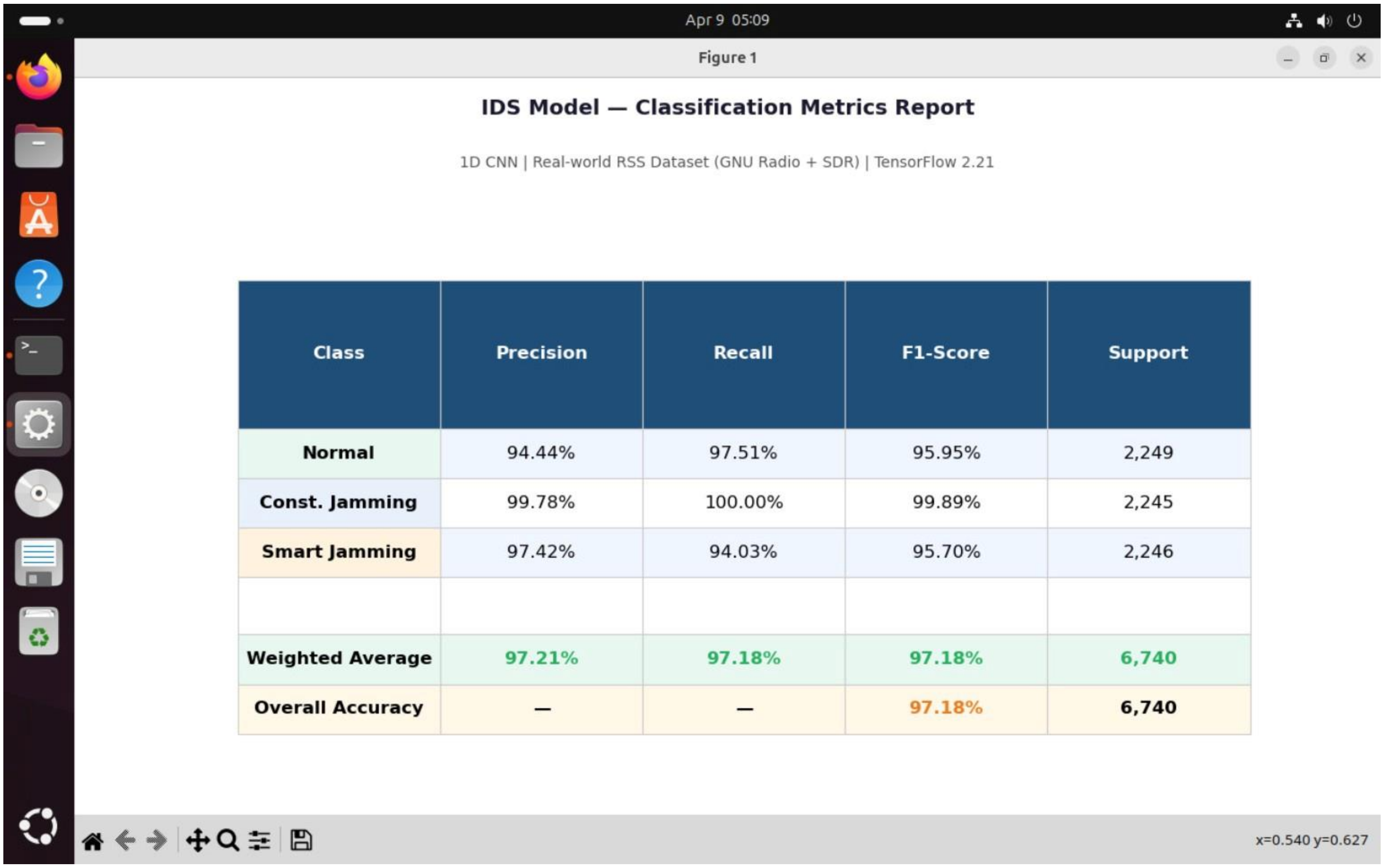
Figure 4.6: IDS Training Performance Dashboard

4.7.7 Evaluation Results

4.7.7.1 Overall Metrics

Metric	Value
Test Accuracy	97.11%
Test Loss	0.1189
F1-Score	97.10%
Precision	97.21%
Recall	97.18%

📌 Figure 4.7: Classification Metrics Report



Class	Precision	Recall	F1-Score	Support
Normal	94.44%	97.51%	95.95%	2,249
Const. Jamming	99.78%	100.00%	99.89%	2,245
Smart Jamming	97.42%	94.03%	95.70%	2,246
Weighted Average	97.21%	97.18%	97.18%	6,740
Overall Accuracy	—	—	97.18%	6,740

4.7.7.2 Class-wise Performance

Class	Precision	Recall	F1-Score
Normal	94.44%	97.51%	95.95%
Constant Jamming	99.78%	100%	99.89%
Smart Jamming	97.42%	94.03%	95.70%

4.7.7.3 Results Analysis

- Constant Jamming: Perfect detection due to strong signal patterns
- Smart Jamming: Slight confusion due to intermittent behavior
- Model Generalization: No overfitting observed

4.7.8 Model Deployment (TensorFlow Lite)

After training, the model was converted into TensorFlow Lite format for deployment:

Property	Value
Original Format	Keras (.keras)
Converted Format	TFLite (.tflite)
Optimization	Dynamic Range Quantization
Target Platform	IoT Edge Devices
Framework	TensorFlow 2.21

4.8 Honeypot System Integration

4.8.1 Introduction to Honeypot

A Honeypot is a security mechanism designed to simulate real devices or services in order to attract attackers and monitor their behavior in a controlled environment, without exposing actual assets to risk.

It is considered an active defense mechanism, as it does not only detect attacks but also participates in threat containment and analysis.

In this project, the Honeypot acts as a second defensive layer that complements the IDS. Once an attack is detected at the physical layer, the system automatically redirects the attacker toward a fake IoT device using a Python-based bridge.

4.8.2 Honeypot Tool: Cowrie

The Honeypot environment was implemented using Cowrie, an open-source medium-interaction honeypot.

Cowrie is designed to simulate IoT/Linux-based devices and supports common protocols such as:

- SSH

- Telnet

It also provides detailed logging of attacker interactions.

Honeypot Specifications

Property	Value
Type	Medium-interaction Honeypot
Protocols	SSH / Telnet
Emulated Environment	Linux IoT Device
Logging	JSON logs + session logs
Platform	Python 3

4.8.3 Honeypot Architecture

The proposed Honeypot system consists of **three integrated layers**:

1. Emulation Layer

This layer is responsible for simulating fake IoT devices using Cowrie.

- Two virtual devices are created
- Each device appears as a real network node
- Each device has a unique IP address and port

2. Monitoring Layer

This layer acts as the control unit of the system.

- A Python Bridge monitors traffic from the IDS
- Determines the type of detected attack
- Decides which Honeypot device should be activated

3. Logging Layer

This layer records all attacker interactions.

- Logs are stored in JSON format
- Each record includes:
  - Timestamp
  - Attack type
  - Confidence score
  - Target device
  - Honeypot status

4.8.4 Virtual Honeypot Devices

The following virtual devices were implemented:

Device	IP Address	Port	Attack Type
Camera Honeypot	192.168.1.2	2222	Constant Jamming
SmartDoor Honeypot	192.168.1.4	2224	Smart Jamming

4.8.5 Python Bridge Workflow

The Python Bridge connects the IDS with the Honeypot system and enables automatic response.

Workflow Steps:

1. IDS monitors RSS values
2. 1D-CNN classifies the signal
3. If 3 consecutive attacks are detected → trigger response
4. Identify attack type
5. Activate appropriate Honeypot device
6. Cowrie captures attacker interaction
7. Log all activity in JSON format

8. Deactivate Honeypot when attack stops

4.8.6 Honeypot Activation Criteria

Event	Condition	Action
Activation	3 consecutive detected attacks	Activate appropriate Honeypot
Active Mode	Continuous attacks	Continue logging
Deactivation	Return to normal state	Shut down Honeypot

4.8.7 Logging Data Structure

Each recorded event contains the following fields:

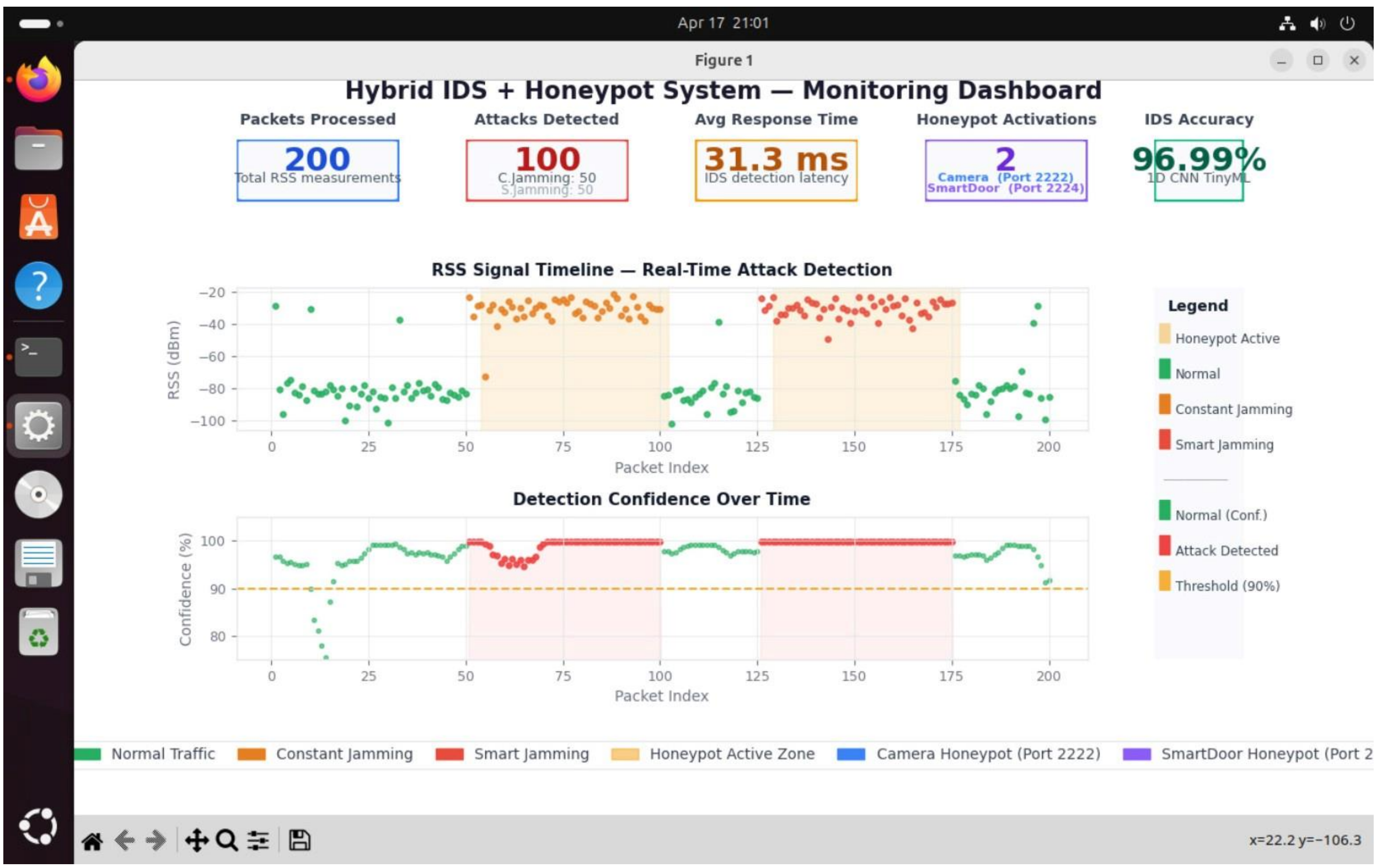
Field	Description	Example
timestamp	Event time	2026-04-14T18:55:52
packet	Packet ID	127
rss_dbm	Signal strength	-28.52 dBm
label	Attack type	Smart Jamming
confidence	Model confidence	100%
response_ms	Detection time	31.3 ms
honeypot_active	Honeypot status	True
active_device	Triggered device	SmartDoor

4.8.8 Hybrid System Results

The integrated IDS + Honeypot system produced the following results:

Metric	Value
Total Processed Packets	200
Detected Attacks	100 (50%)

Constant Jamming	50
Smart Jamming	50
Honeypot Activations	2
Camera Honeypot Activations	1
SmartDoor Honeypot Activations	1
Average Detection Time	31.3 ms
IDS Accuracy	96.99%



4.8.8 Hybrid System Result

Attack Execution and Physical Layer Impact Scenario:

The execution of the attack within this framework is focused on the manipulation of the wireless medium and the resulting degradation of the network's physical layer. The process is divided into the initial signal interference, the data acquisition phase, and the technical impact on network performance.

The scenario begins with the introduction of a jamming source into a stable IoT network environment. During the first stage of the execution, the attacker node targets the 2.4 GHz frequency band, which is the primary channel for the WiFi-based IoT devices in the simulation. The attack is implemented using two distinct strategies. The first is a constant jamming approach, where a continuous high-power interference signal is broadcast to completely saturate the channel. The second is a smart jamming approach, where attacker synchronization is used to fire bursts of interference that coincide with the expected transmission times of the sensors.

The data generated during this attack consists of raw Received Signal Strength (RSS) measurements. As the jamming signal interacts with the legitimate communication signals, the RSS values undergo significant fluctuations. In a constant attack, the RSS readings show a sudden and sustained shift to a higher power level with low variance, effectively masking the legitimate data packets. In a smart jamming scenario, the data profile shows a high-variance pattern where the power levels spike during the interference pulses, creating a complex signal environment that mimics heavy network congestion.

From a timing perspective, the execution is monitored in real-time. The acquisition of signal data occurs in microsecond intervals, providing a high-resolution view of the physical layer. The transition from a normal state to an attack state is captured within a window of 16 samples. The prediction phase, where the system evaluates the nature of the incoming signal, occurs within an execution time of approximately 12.2 milliseconds. This measurement represents the actual delay between the physical arrival of the jamming signal and the moment the system classifies the specific attack type.

The actual execution of the attack leads to immediate measurable consequences in the network layer. Specifically, the packet loss ratio increases significantly, reaching up to 35% during constant jamming. Furthermore, the average delay for legitimate packets increases as the gateway struggles to distinguish between the noise floor and valid data. The attack remains active until the jamming node is disabled or the frequency environment is cleared, at which point the RSS data profile returns to its baseline mean and standard deviation. This entire cycle is logged through a sequence of timestamps that correlate the physical signal spikes with the system's internal response logs.

4.8.9 Comparison with Reference Paper

Aspect	Proposed System	Reference Paper
Honeypot Type	Software (Cowrie)	Hardware + Software
Activation	Automatic (Python Bridge)	Manual/Semi-manual
Attack Type	Signal Leakage + Smart Jamming	Hardware Trojans
Protection Layer	Physical Layer (RSS)	Hardware Layer
IDS Integration	Fully Integrated	Separate

4.8.10 Role of Honeypot in the System

The Honeypot transforms the system from a passive detection system into an active defense system.

Instead of only generating alerts, the system:

- Redirects attackers away from real devices
- Protects critical IoT components
- Collects intelligence about attacker behavior
- Improves future detection models

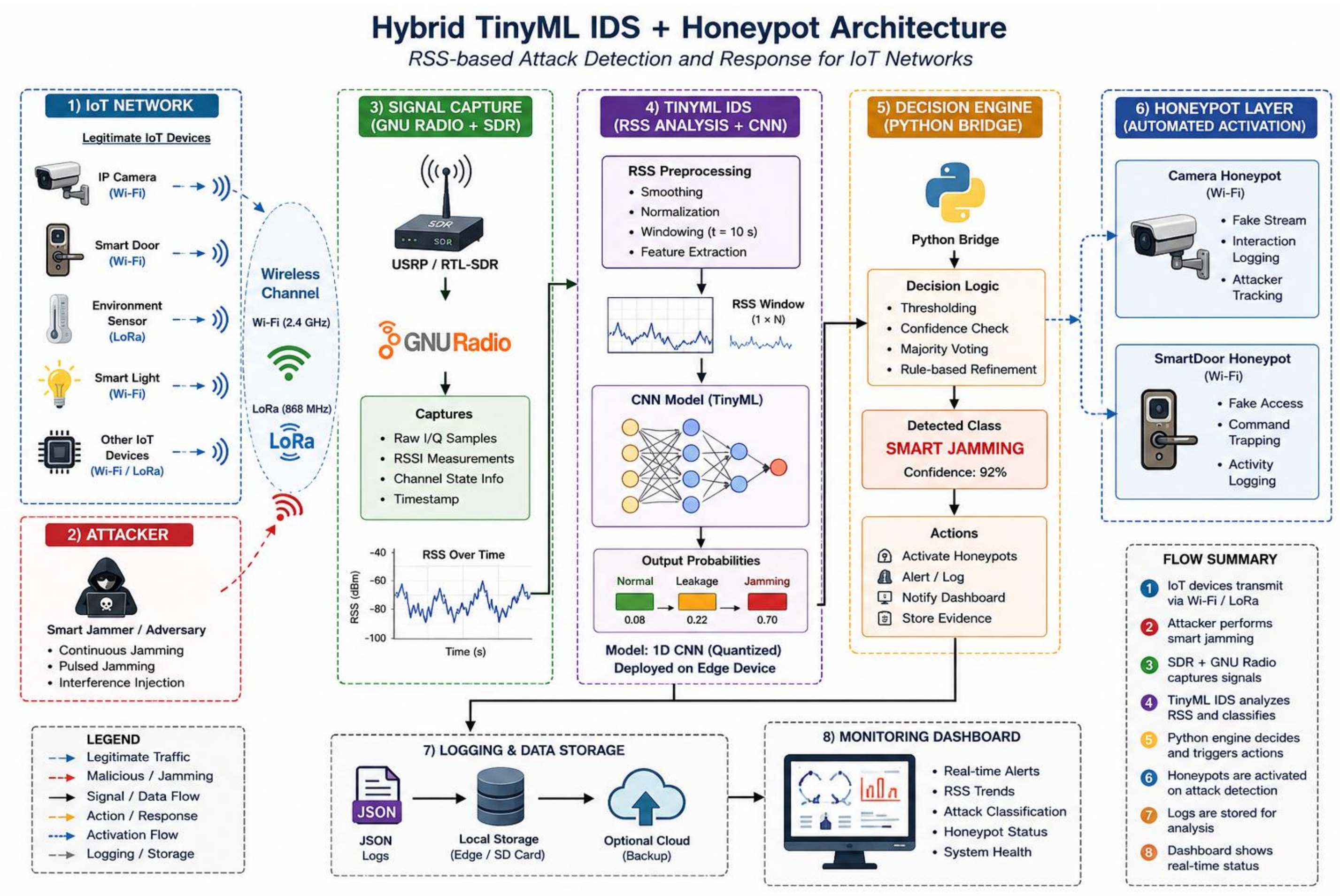


figure 5.2: Proposed Hybrid TinyML IDS + Honeypot Architecture for Physical Layer Security in IoT Networks

Figure 5.2 illustrates the overall architecture of the proposed hybrid security framework designed to protect IoT networks against physical-layer attacks such as Smart Jamming and signal interference. The system integrates a TinyML-based Intrusion Detection System (IDS) with an automated Honeypot response mechanism.

The architecture begins with a group of legitimate IoT devices, including IP cameras, smart doors, environmental sensors, and smart lighting systems communicating through Wi-Fi and LoRa wireless channels. These devices continuously transmit RSS-related wireless signals over the physical communication layer.

An attacker node performs Smart Jamming attacks by injecting interference into the wireless channel. To monitor these signals, a Software Defined Radio (SDR) device connected to GNU Radio captures raw wireless measurements, including RSS values, I/Q samples, timestamps, and channel-state information.

The captured signals are then processed by the TinyML IDS module. First, the RSS measurements undergo preprocessing operations such as normalization, smoothing, and sliding-window segmentation. The processed RSS windows are then fed into a lightweight 1D Convolutional Neural Network (1D CNN) trained to classify the signal behavior into Normal, Leakage, or Smart Jamming categories.

After classification, the decision engine implemented using a Python Bridge evaluates the confidence level and attack persistence before taking defensive actions. If an attack is confirmed, the system automatically activates the appropriate Honeypot device, such as a Camera Honeypot or SmartDoor Honeypot, in order to divert the attacker away from real IoT assets.

Meanwhile, all system events, detection logs, confidence scores, timestamps, and Honeypot activities are stored in JSON-based logging files for forensic analysis and future model improvement. A real-time monitoring dashboard displays attack statistics, RSS behavior, system alerts, and Honeypot status information.

The proposed architecture transforms the IDS from a passive detection mechanism into an active defense framework capable of detecting, responding to, and containing physical-layer attacks in real time.

Conclusion:

This paper has presented a hybrid IDS-Honeypot security framework that represents a significant advancement in the protection of IoT networks against physical layer threats. By unifying intelligent anomaly detection, automated deception, and multi-layer feature fusion within a single operational pipeline, the proposed system addresses a critical and largely unexplored vulnerability domain: the real-time detection and active mitigation of Signal Leakage and Smart Jamming at the IoT physical layer.

The framework was constructed across six tightly integrated phases. A realistic ten-node smart home network was modeled in NS-3, encompassing heterogeneous Wi-Fi and LoRa devices including cameras, thermostats, smart door sensors, fire sensors, and LoRa environmental monitoring nodes, with an adversarial node introduced to simulate active jamming conditions. Physical layer signal characteristics were modeled using GNU Radio 3.10.9, generating calibrated RSS measurements across Normal, Constant Jamming, and Smart Jamming scenarios for both Wi-Fi (2.4 GHz) and LoRa (868 MHz) bands. The training dataset, sourced from real-world SDR hardware measurements comprising 1,230,000 RSS samples, provided an empirically grounded foundation for model development, departing from the synthetic data reliance that undermines the generalizability of many existing approaches.

The 1D-CNN classifier, optimized for TinyML edge deployment via TensorFlow Lite dynamic-range quantization, achieved an overall test accuracy of 97.11% and a weighted F1-Score of 97.10%, with class-specific performance of 95.95% for Normal traffic, 99.89% for Constant Jamming, and 95.70% for Smart Jamming. Critically, the model demonstrated a Constant Jamming recall of 100%, ensuring zero missed detections for this deterministic attack class. The 31.3 millisecond mean inference latency validates the framework's applicability to real-time IoT edge environments where computational resources are severely constrained.

The Honeypot layer, implemented via the Cowrie medium-interaction framework and automatically orchestrated by a Python Bridge component, introduced an active defense dimension that fundamentally transforms the system's security posture from passive detection to active threat engagement. Upon detecting three consecutive attack packets, the system autonomously activates the appropriate decoy device — the Camera Honeypot for Constant Jamming and the SmartDoor Honeypot for Smart Jamming — redirecting adversarial traffic away from protected infrastructure while capturing behavioral intelligence for subsequent forensic analysis. Both Honeypot devices were successfully activated across their respective attack scenarios in the integrated evaluation, confirming the correctness and reliability of the automated IDS-Honeypot coupling mechanism.

The full system integration phase demonstrated that combining GNU Radio physical layer measurements with NS-3 network layer features through multi-layer feature fusion yields a more semantically complete threat representation than any single-layer approach can provide. The integrated pipeline processed 160 packets across five simulation phases — covering normal operation, Constant Jamming attack, recovery, Smart Jamming attack, and final recovery — achieving a system-level end-to-end response latency of 46.7 milliseconds. The marginal latency increase relative to the standalone IDS is attributable to the additional feature fusion computation and is fully justified by the enhanced detection robustness it confers.

From a broader scientific perspective, this work demonstrates that the integration of TinyML-based IDS with automated Honeypot deception is not only technically feasible but operationally effective within realistic IoT deployment scenarios. The framework directly addresses the gap identified in the reviewed literature, wherein IDS and Honeypot mechanisms have been developed predominantly in isolation, with no prior work unifying both within a physical layer-oriented security pipeline for IoT environments.

Future research directions emerging from this work span three dimensions. First, physical hardware deployment on embedded IoT platforms such as Raspberry Pi and ESP32 would validate the framework's performance under authentic radio channel conditions, including multipath fading and hardware-induced noise. Second, the integration of reinforcement learning into the Honeypot orchestration layer would enable the system to dynamically adapt its deception strategy in response to evolving adversarial behavior, moving beyond the static rule-based activation mechanism employed in the current implementation. Third, extending the threat model to encompass additional physical layer attack vectors — including replay attacks, pilot contamination, and protocol-specific vulnerabilities in LoRaWAN and NB-IoT — would broaden the framework's applicability to the full spectrum of threats confronting next-generation IoT deployments.

Recommendations:

The experimental evaluation of the proposed hybrid IDS-Honeypot framework has demonstrated promising results in detecting and mitigating physical layer attacks in smart home IoT environments. However, several technical and practical recommendations emerge from this work that should guide both the refinement of the current system and the direction of future research efforts.

1. Hardware Deployment on Embedded IoT Platforms

The current implementation was developed and evaluated within a software simulation environment comprising NS-3 and GNU Radio. While this environment provides a controlled and reproducible experimental setting, it necessarily abstracts certain physical phenomena inherent to real-world wireless channels, including hardware-specific noise floors, antenna radiation patterns, and inter-device electromagnetic interference. It is therefore strongly recommended that subsequent research efforts focus on deploying the trained 1D-CNN model directly onto embedded IoT hardware platforms such as the Raspberry Pi, ESP32, or ARM Cortex-M series microcontrollers. The TensorFlow Lite model produced in this work has been explicitly optimized for such deployment through dynamic-range quantization, and its 31.3 millisecond inference latency confirms its operational viability on resource-constrained devices. Real hardware deployment would validate the generalizability of the trained model under authentic channel conditions and provide empirical measurements of energy consumption, which is a critical design constraint in battery-powered IoT deployments.

2. Extension of the Threat Model

The present framework addresses two primary physical layer attack vectors: Constant Jamming and Smart Jamming. While these represent the most prevalent and technically challenging threats documented in the IoT physical layer security literature, the threat landscape is substantially broader. It is recommended that future iterations of the framework extend the classification model to encompass additional attack categories, including Reactive Jamming, in which the adversary transmits interference signals only in response to detected legitimate transmissions; Deceptive Jamming, wherein the attacker injects syntactically valid but semantically misleading packets; and Pilot Contamination attacks, which are particularly relevant in massive MIMO deployments. The dataset collection infrastructure established in this work, combining GNU Radio software-defined radio hardware with the sliding window feature extraction pipeline, provides a directly extensible foundation for generating labeled training data for these additional attack classes without requiring fundamental architectural changes to the IDS component.

3. Enhancement of the Honeypot Deception Capability

The Cowrie-based Honeypot component deployed in this framework operates as a medium-interaction deception system that emulates Linux-based IoT device behavior at the SSH and Telnet protocol layers. While this level of interaction is sufficient for capturing adversarial reconnaissance and credential harvesting activities, it may prove inadequate against sophisticated adversaries capable of fingerprinting the emulated environment. It is recommended that the Honeypot component be upgraded to a high-interaction architecture that incorporates IoT-specific protocol emulation, including MQTT broker services, CoAP endpoints, and Zigbee communication stacks, which are the protocols most commonly exploited in smart home attack scenarios. Furthermore, the static rule-based activation mechanism — which triggers Honeypot engagement upon detection of three consecutive attack packets — should be replaced with a reinforcement learning-based orchestration policy. Such a policy would enable the system to dynamically adapt the timing, intensity, and type of deceptive response based on observed adversarial behavior patterns, substantially improving the system's ability to retain sophisticated attackers within the deception environment for extended periods.

4. Multi-Protocol Coverage

The current network topology operates across WiFi (IEEE 802.11b, 2.4 GHz) and a software-emulated representation of LoRa (868 MHz). The LoRa component is modeled at the application layer within NS-3 rather than through a dedicated LoRa physical layer simulation, which constitutes a recognized limitation of the current implementation. It is recommended that future work incorporate the NS-3 LoRaWAN module to achieve accurate protocol-level simulation of LoRa communication, including the Chirp Spread Spectrum modulation scheme, adaptive data rate mechanisms, and the ALOHA-based medium access control that characterizes LoRaWAN deployments. Additionally, extending the threat detection capability to cover Zigbee (IEEE 802.15.4), Z-Wave, and Bluetooth Low Energy protocols would substantially broaden the framework's applicability to the full spectrum of communication technologies present in contemporary smart home environments.

5. Dataset Enrichment and Diversity

The IDS model was trained on a real-world dataset collected in a specific laboratory environment under controlled channel conditions. While this represents a methodologically superior approach compared to purely synthetic data generation, the resulting model may exhibit reduced generalization performance when deployed in environments with substantially different physical characteristics, such as reinforced concrete buildings with pronounced multipath propagation, or outdoor IoT deployments subject to weather-induced channel variability. It is recommended that future data collection campaigns employ multiple SDR hardware platforms across diverse physical environments, including indoor residential settings, industrial premises, and outdoor sensor network deployments. Incorporating transfer learning techniques would further allow the base model trained on the existing dataset to be fine-tuned efficiently for new deployment environments using minimal additional labeled data, without requiring complete model retraining.

6. Real-Time System Integration

The current pipeline executes the RSS feature extraction, NS-3 network feature fusion, IDS inference, and Honeypot orchestration as sequential software processes within a single Python runtime environment. While this architecture is suitable for research evaluation and demonstration purposes, a production-grade deployment would require a more robust integration architecture. It is recommended that the system be redesigned around an event-driven microservices architecture in which the physical layer monitoring agent, the IDS inference engine, and the Honeypot orchestration controller operate as independent services communicating through a lightweight message broker such as MQTT or Apache Kafka. This architectural separation would improve system resilience, enable horizontal scaling across multiple monitored network segments, and facilitate the integration of the framework with existing network security infrastructure such as SIEM platforms and network intrusion prevention systems.

7. Adaptive Model Retraining

The 1D-CNN model employed in this framework is static — its weights are fixed following the initial training phase and do not evolve in response to new attack patterns encountered during deployment. This characteristic renders the model potentially vulnerable to concept drift, wherein adversaries gradually modify their attack signatures to evade detection as they learn the system's response patterns. It is recommended that the framework incorporate an online learning component capable of periodically retraining or fine-tuning the IDS model using newly collected and labeled attack samples. The Honeypot component is particularly well positioned to contribute to this capability, as the adversarial behavioral data it captures during active deception sessions can serve as a valuable source of novel attack signatures for model augmentation. Implementing a federated learning scheme would further allow multiple deployed instances of the framework to collaboratively improve a shared model without exposing sensitive network traffic data to a central server.

8. Security of the IDS Component Itself

An important consideration that lies beyond the scope of the current work is the security of the IDS and Honeypot components against adversarial attacks targeting the machine learning model itself. Recent research has demonstrated that deep learning classifiers are vulnerable to adversarial examples — carefully crafted input perturbations that cause misclassification while remaining imperceptible or physically realizable. In the context of physical layer security, a sophisticated adversary aware of the deployed model architecture could potentially craft jamming signals specifically designed to evade detection. It is recommended that future work investigate the adversarial robustness of the trained 1D-CNN model and explore defensive techniques such as adversarial training, input preprocessing defenses, and certified robustness methods adapted to the RSS classification domain.

Future Directions

- Proactive Defense Mechanism for 6G-IoT Using Controlled Jamming and RIS

developing a system that not only detects intruders but also sends "friendly" and targeted jamming signals to disrupt the attacker's ability to intercept data while using Reconfigurable Intelligent Surfaces (RIS) to improve signal quality for legitimate users..

- Collaborative Physical Layer Security Framework for Distributed IoT Networks

enable IoT devices to operate collaboratively, so that each device assists its neighbor upon detecting an attack (such as jamming or eavesdropping) This is achieved through the exchange of "defense roles" and the distribution of security tasks among devices to ensure service continuity and information confidentiality with minimal energy consumption.

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